❖ DEFİNİTİON OF THE ALGORİTHM

The algorithm performs n-gram analysis and constructs Markov models for textual data. Below is a detailed definition of the algorithm, including its functions, procedures, inputs, and outputs.

1. Algorithm Workflow

Text Cleaning:

- Prepares the text by retaining only letters and spaces, and converting all text to lowercase.

- Converts all characters to lowercase.

- Removes all non-letter characters except whitespace.

- Replaces newline (`\n`) and carriage return (`\r`) characters with spaces.

- Trims leading and trailing spaces.

Unigram Analysis:

- Calculates word frequencies and their probabilities in the text.

N-Gram Analysis:

- Constructs prefix-next word pairs for the given `n` and computes transition probabilities.

Result Display:

- Displays the results of the unigram analysis and the top 50 probabilities for the 2-gram and 3-gram Markov models.

2. Main Input and Output

- **Input:**

  - A text file containing the text to be analyzed (specified in the `filePath` variable).

- **Output:**

- 1-gram frequency analysis (word counts and probabilities).

- 2-gram and 3-gram Markov model probabilities, displaying the most likely transitions.

3. Functions and Procedures

```java
public class nGramAnalysis {
    public static void main(String[] args) throws IOException {
        String filePath = "C:\\Users\\huvey\\Downloads\\Novel-Samples\\DENEMELER.TXT"; // Put the text file to be tested here
        String text = readFile(filePath);

        // Cleaning process: Leave only letters and residue
        StringBuilder cleanedText = new StringBuilder();
        for (char c : text.toCharArray()) {
            if (Character.isLetter(c) || Character.isWhitespace(c)) {
                cleanedText.append(Character.toLowerCase(c));
            }
        }

        String cleaned = cleanedText.toString()
                .replace("\r", " ") //  clear `\r` characters
                .replace("\n", " ") // clear `\n` characters
                .trim(); // Clear leading and trailing spaces


        // 1-gram analysis
        System.out.println("\n1-Gram Analizi:");
        unigramAnalysis(cleaned); // Use cleaned text

        // 2-gram and 3-gram analysis available for Markov model
        System.out.println("\n2-Gram Markov Model:");
        buildMarkovModel(cleaned, 1); // Use cleaned text

        System.out.println("\n3-Gram Markov Model:");
        buildMarkovModel(cleaned, 2); // Use cleaned text
    }
```

`main(String[] args)` 1

`main(String[] args)`

- **Purpose:**

  - Coordinates the n-gram analysis by calling other functions.

- **Inputs:**

  - File path of the text to be analyzed.

- **Outputs:**

  - Results of unigram analysis and Markov models (2-gram and 3-gram) displayed on the console.

```java
    // Read text from file
    private static String readFile(String filePath) throws IOException {
        StringBuilder sb = new StringBuilder();
        try (BufferedReader br = new BufferedReader(new InputStreamReader(new FileInputStream(filePath), "UTF-8"))) {
            String line;
            while ((line = br.readLine()) != null) {
                sb.append(line).append(" ");
            }
        }
        return sb.toString();
    }
```

`readFile(String filePath)` 1

`readFile(String filePath)`

- **Purpose:**

- Reads the content of a text file and returns it as a single string.

- **Inputs:**

- `filePath` (String): Path to the input text file.

- **Outputs:**

- `text` (String): Full content of the file as a single string.

```java
// 1-Gram Analysis: Calculate the frequency of words
private static void unigramAnalysis(String text) {
    String[] words = text.split("\\s+");
    Map<String, Integer> wordCounts = new HashMap<>();

    // Calculate word frequencies
    for (String word : words) {
        wordCounts.put(word, wordCounts.getOrDefault(word, 0) + 1);
    }

    // Find total word count
    int totalWords = Arrays.stream(words).mapToInt(w -> 1).sum();

    // Print word frequencies and probabilities
    System.out.println("Word Frequencies and Probabilities (Top 50):");
    wordCounts.entrySet().stream()
            .sorted((a, b) -> b.getValue().compareTo(a.getValue())) // Frekansa göre azalan sırada
            .limit(50) // İlk 50'yi seç
            .forEach(entry -> {
                String word = entry.getKey();
                int count = entry.getValue();
                double probability = (double) count / totalWords;
                System.out.printf("%s: %d (%.4f)%n", word, count, probability);
            });
}
```

`unigramAnalysis(String text)` 1

`unigramAnalysis(String text)`

- **Purpose:**

  - Performs a frequency analysis of individual words (1-gram analysis) in the cleaned text.

- **Inputs:**

  - `text` (String): The cleaned text string.

- **Procedure:**

  1. Splits the text into words using whitespace as the delimiter.
  2. Counts the frequency of each word using a `HashMap`.
  3. Computes the total number of words.
  4. Calculates the probability of each word as `frequency / total words`.

- **Outputs:**

  - Prints the top 50 words with their frequencies and probabilities.

```java
    // N-Gram Analysis for Markov model
    private static void buildMarkovModel(String text, int n) {
        String[] words = text.split("\\s+");
        Map<String, Map<String, Integer>> markovModel = new HashMap<>();

        // N-gram creation
        for (int i = 0; i <= words.length - n - 1; i++) {
            String prefix = String.join(" ", Arrays.copyOfRange(words, i, i + n)); // Prefix
            String nextWord = words[i + n]; // Next word

            markovModel.putIfAbsent(prefix, new HashMap<>());
            Map<String, Integer> nextWordCounts = markovModel.get(prefix);

            // Increase crossover frequency
            nextWordCounts.put(nextWord, nextWordCounts.getOrDefault(nextWord, 0) + 1);
        }

        // Probability calculations and ranking
        List<Map.Entry<String, Double>> probabilities = new ArrayList<>();
        for (Map.Entry<String, Map<String, Integer>> entry : markovModel.entrySet()) {
            String prefix = entry.getKey();
            Map<String, Integer> nextWords = entry.getValue();
            int total = nextWords.values().stream().mapToInt(Integer::intValue).sum();

            for (Map.Entry<String, Integer> wordEntry : nextWords.entrySet()) {
                String word = wordEntry.getKey();
                int count = wordEntry.getValue();
                double probability = (double) count / total;

                probabilities.add(new AbstractMap.SimpleEntry<>(
                        prefix + " -> " + word, probability
                ));
            }
        }
```

`buildMarkovModel(String text, int n)` 1

```java
// Sort the possibilities in descending order
probabilities = probabilities.stream()
        .sorted((a, b) -> Double.compare(b.getValue(), a.getValue()))
        .collect(Collectors.toList());

// Print first 50 results
System.out.println("\n50 Highest Probabilities:");
for (int i = 0; i < Math.min(100, probabilities.size()); i++) {
    Map.Entry<String, Double> entry = probabilities.get(i);
    System.out.printf("%d. %s: %.4f%n", i + 1, entry.getKey(), entry.getValue());
}
```

`buildMarkovModel(String text, int n)` 2

`buildMarkovModel(String text, int n)`

- **Purpose:**

- Constructs an n-gram-based Markov model and calculates transition probabilities for `n+1`-grams.

- **Inputs:**

- `text` (String): The cleaned text string.

- `n` (int): The size of the n-gram prefix (e.g., `n=1` for a 2-gram model).

- **Procedure:**

  1. Splits the text into words.
  2. Iterates over the text to create prefix-next word pairs.
  3. Uses a nested `HashMap` to store frequencies of prefix-next word pairs.
  4. Computes the probability of each pair as `count / total occurrences of prefix`.
  5. Sorts and ranks the pairs by their probabilities in descending order.

- **Outputs:**

  - Prints the top 50 most probable transitions (e.g., `prefix → next word`) with their probabilities.

4. Test Results

Input:

File Name: DENEMELER.txt

Total Running Time: 449 millisecond

```
Total Run Time: 449 millisecond
```

Output:

Top 50 items analyzed for 1-gram Analysis:

bir: 1510 (0,0300)

ve: 895 (0,0178)

bu: 569 (0,0113)

de: 478 (0,0095)

daha: 440 (0,0087)

da: 432 (0,0086)

ki: 399 (0,0079)

iin: 392 (0,0078)

ne: 321 (0,0064)

her: 318 (0,0063)

kadar: 309 (0,0061)

o: 301 (0,0060)

en: 285 (0,0057)

kendi: 253 (0,0050)

gibi: 248 (0,0049)

ama: 233 (0,0046)

ok: 231 (0,0046)

kitap: 192 (0,0038)

btn: 182 (0,0036)

blm: 179 (0,0036)

yle: 176 (0,0035)

insan: 173 (0,0034)

zaman: 171 (0,0034)

baka: 159 (0,0032)

deil: 156 (0,0031)

byk: 151 (0,0030)

ey: 147 (0,0029)

nk: 146 (0,0029)

olan: 143 (0,0028)

iyi: 131 (0,0026)

hi: 127 (0,0025)

hibir: 121 (0,0024)

ben: 116 (0,0023)

bile: 115 (0,0023)

bana: 114 (0,0023)

onu: 114 (0,0023)

bizim: 112 (0,0022)

sonra: 112 (0,0022)

onun: 111 (0,0022)

yalnz: 103 (0,0020)

pek: 102 (0,0020)

kendini: 100 (0,0020)

gre: 95 (0,0019)

az: 94 (0,0019)

lm: 94 (0,0019)

ona: 94 (0,0019)

diye: 92 (0,0018)

ya: 91 (0,0018)

benim: 89 (0,0018)

bizi: 89 (0,0018)

Top 50 items analyzed for 2-gram Analysis:

1. gnlerinizden -> daha: 1,0000
2. tenebrasque -> resolvit: 1,0000
3. eidi -> vardr: 1,0000
4. fulget -> honoribus: 1,0000
5. zevksizlik -> yznden: 1,0000
6. yaamlar -> var: 1,0000
7. geirmiti -> lecei: 1,0000
8. kralna -> hizmet: 1,0000
9. kalmadn -> nk: 1,0000
10. zamankinden -> baka: 1,0000
11. kalmadm -> gerekten: 1,0000
12. hastalklardan -> dinlere: 1,0000
13. azmzn -> tadyla: 1,0000
14. grnndeki -> yeri: 1,0000
15. gideriz -> nsan: 1,0000
16. dzeltir -> ceza: 1,0000
17. vermesini -> bildim: 1,0000
18. saygya -> gelince: 1,0000
19. dkerken -> ve: 1,0000
20. giderin -> yarn: 1,0000
21. giderim -> hey: 1,0000
22. sayanlar -> bre: 1,0000
23. grmeyi -> hie: 1,0000

24. bakasnz -> diye: 1,0000
25. bulanlar -> bile: 1,0000
26. beendiiniz -> yapsndaki: 1,0000
27. gzlleri -> topallar: 1,0000
28. korkusunda -> olursak: 1,0000
29. giriir -> ve: 1,0000
30. plenus -> vitae: 1,0000
31. czaml -> olmaya: 1,0000
32. uzandaki -> tank: 1,0000
33. mayanzdadr -> ondan: 1,0000
34. zavalls -> en: 1,0000
35. rtyle -> kaparlarm: 1,0000
36. trelerimizi -> bizim: 1,0000
37. psrklk -> korkaklksa: 1,0000
38. bildim -> takn: 1,0000
39. filozoftu -> ki: 1,0000
40. hekimlikteki -> eski: 1,0000
41. yabanclardan -> ve: 1,0000
42. kayglarndan -> kurtulmu: 1,0000
43. ertesi -> gn: 1,0000
44. atmamalyz -> nk: 1,0000
45. arzulanan -> bir: 1,0000
46. drtkleyen -> kitap: 1,0000
47. srnn -> oyuna: 1,0000
48. netameli -> daha: 1,0000
49. r -> stne: 1,0000
50. terlermi -> byle: 1,0000

Top 50 items analyzed for 3-gram Analysis:

1. ki skenderin -> yan: 1,0000
2. eyler diler -> bunlarn: 1,0000
3. alabileceini sand -> gibi: 1,0000

4. aknlktan dona -> kalp: 1,0000
5. genken biberli -> baharl: 1,0000
6. kendimden ok -> ona: 1,0000
7. nec superstes -> integer: 1,0000
8. aka ortaya -> koyar: 1,0000
9. ovduruyor sardryor -> ve: 1,0000
10. evinde kald -> bir: 1,0000
11. lakedemonyal kadnlar -> ki: 1,0000
12. atmalarn hibiri -> insanla: 1,0000
13. amplius nostrum -> est: 1,0000
14. biroklar da -> vardr: 1,0000
15. qui cupit -> aut: 1,0000
16. eskrimde stnlk -> korkak: 1,0000
17. hekimlerle alayp -> szlayanlarla: 1,0000
18. birka harf -> antiokusa: 1,0000
19. duada kendileri -> iin: 1,0000
20. niteliine uygun -> olarak: 1,0000
21. bakmndan onlarn -> zaferleri: 1,0000
22. kalabalktan kamak -> yetmez: 1,0000
23. verebileceini sand -> srece: 1,0000
24. gzellik aklmz -> iletmez: 1,0000
25. nandrmann yeni -> bir: 1,0000
26. sere yuvasn -> hi: 1,0000
27. tadn ve -> rengini: 1,0000
28. birbirimize ekecek -> hatrlatacak: 1,0000
29. dnceleri ileri -> geri: 1,0000
30. rlplak da -> gsterirdim: 1,0000
31. properandus et -> acri: 1,0000
32. tutmak isteyen -> lakedemonyallar: 1,0000
33. dertli clz -> insanolu: 1,0000
34. da mideyi -> bozar: 1,0000
35. olmay yeter -> saym: 1,0000
36. olan acaba -> hangimizin: 1,0000

37. bulanmasnn nedeni -> stnde: 1,0000

38. bizde pek -> fazla: 1,0000

39. ok vermekte -> kendini: 1,0000

40. hep yer -> deitiriyor: 1,0000

41. zaman aclarm -> trplerler: 1,0000

42. akn gevememesi -> iin: 1,0000

43. olmal ama -> mutluluunu: 1,0000

44. vermeden salanamaz -> yle: 1,0000

45. nasl veriz -> hzl: 1,0000

46. bulants geivermi -> senecann: 1,0000

47. dehetiyle allak -> bullak: 1,0000

48. benzeri stnlkler -> gibi: 1,0000

49. yoktur ekip -> gitmezseniz: 1,0000

50. komutanlar trl -> halleriyle: 1,0000

Input:

File Name: BİLİM İŞ BAŞINDA.txt

Total Running Time: 295 millisecond

```
Total Run Time: 295 millisecond
```

Output:

Top 50 items analyzed for 1-gram Analysis:

bir: 454 (0,0490)

ve: 202 (0,0218)

bu: 154 (0,0166)

iin: 91 (0,0098)

de: 77 (0,0083)

ok: 74 (0,0080)

ile: 71 (0,0077)

daha: 66 (0,0071)

da: 63 (0,0068)

olarak: 52 (0,0056)

ancak: 50 (0,0054)

elektrik: 47 (0,0051)

veya: 43 (0,0046)

su: 43 (0,0046)

kadar: 41 (0,0044)

byk: 37 (0,0040)

gibi: 30 (0,0032)

biimde: 30 (0,0032)

enerji: 29 (0,0031)

olan: 29 (0,0031)

iki: 29 (0,0031)

doru: 27 (0,0029)

yaklak: 24 (0,0026)

yk: 24 (0,0026)

pek: 24 (0,0026)

zaman: 24 (0,0026)

oysa: 22 (0,0024)

dier: 22 (0,0024)

iyi: 22 (0,0024)

ne: 21 (0,0023)

deniz: 20 (0,0022)

zerinde: 20 (0,0022)

olduu: 20 (0,0022)

sonra: 20 (0,0022)

ilk: 20 (0,0022)

nce: 19 (0,0020)

cam: 19 (0,0020)

en: 18 (0,0019)

her: 18 (0,0019)

biri: 17 (0,0018)

basn: 17 (0,0018)

sa: 17 (0,0018)

suyun: 17 (0,0018)

o: 16 (0,0017)

ey: 16 (0,0017)

kk: 16 (0,0017)

bilim: 16 (0,0017)

olduunu: 16 (0,0017)

geri: 16 (0,0017)

sol: 16 (0,0017)

Top 50 items analyzed for 2-gram Analysis:

1. edilerek -> yapld: 1,0000
2. koyulup -> stlrsa: 1,0000
3. ilemidir -> bir: 1,0000
4. ieren -> bir: 1,0000
5. bebeklerini -> sol: 1,0000
6. paralanr -> ivi: 1,0000
7. herkes -> bilir: 1,0000
8. ten -> ddn: 1,0000
9. artmas -> gerekir: 1,0000
10. besin -> retiminde: 1,0000
11. niversitedeki -> zel: 1,0000
12. cde -> ok: 1,0000
13. gevek -> yn: 1,0000
14. itinden -> bile: 1,0000
15. kvrandka -> bisiklet: 1,0000
16. iitilmeyenler -> daha: 1,0000
17. yangnlara -> veya: 1,0000
18. hakl -> olabilirdi: 1,0000
19. size -> bakacak: 1,0000
20. saraylarda -> kullandklar: 1,0000
21. gllerin -> dipten: 1,0000
22. tanmlamak -> yollardan: 1,0000
23. saklandn -> aklamadan: 1,0000

24. kum -> zerindeki: 1,0000
25. tahtalardan -> gelen: 1,0000
26. scaklklarnn -> sonunda: 1,0000
27. borular -> ve: 1,0000
28. varln -> ortaya: 1,0000
29. grevlisi -> hem: 1,0000
30. elleriyle -> veya: 1,0000
31. olmaktan -> kmt: 1,0000
32. felsefesi -> profesr: 1,0000
33. amzn -> yenilik: 1,0000
34. zorlanrlarsa -> nc: 1,0000
35. abalarndan -> ortaya: 1,0000
36. dolgu -> yol: 1,0000
37. pedallar -> dorudan: 1,0000
38. f -> p: 1,0000
39. atlama -> olasl: 1,0000
40. h -> a: 1,0000
41. edebiyatn -> veya: 1,0000
42. yaplardr -> mermer: 1,0000
43. yknn -> bir: 1,0000
44. bildii -> bir: 1,0000
45. srmt -> simyaclar: 1,0000
46. eilimini -> byk: 1,0000
47. sala -> zarar: 1,0000
48. yaklszlktan -> kvrandka: 1,0000
49. incin -> cm: 1,0000
50. iletken -> bir: 1,0000

Top 50 items analyzed for 3-gram Analysis:

1. sesler kullanlarak -> yaplan: 1,0000
2. ile incelenmesi -> atomlarn: 1,0000
3. deiikliin aslnda -> ayna: 1,0000

4. srd dier -> bir: 1,0000
5. civarnda olan -> iitme: 1,0000
6. felaketleri yaratan -> etken: 1,0000
7. oluabilen elektrik -> ykleri: 1,0000
8. kadar esnek -> hale: 1,0000
9. kuvvet uygular -> kahramanmzn: 1,0000
10. formlde e -> m: 1,0000
11. alabilir bu -> etkileyici: 1,0000
12. paristeki bir -> maazann: 1,0000
13. ve soukluk -> derecelerini: 1,0000
14. uygulanmasna iyi -> bir: 1,0000
15. elde edilebilecek -> enerji: 1,0000
16. yere gitmek -> sz: 1,0000
17. diliyle syleyecek -> olursak: 1,0000
18. suya ilk -> buharlama: 1,0000
19. bir ltmz -> yok: 1,0000
20. eski moda -> duraan: 1,0000
21. gelindiinde alar -> boyunca: 1,0000
22. romanc lord -> snow: 1,0000
23. elektriin ilk -> kurbanlarndan: 1,0000
24. yorumlarnn yer -> ald: 1,0000
25. uygulanmad srece -> olduka: 1,0000
26. elenceli olduunu -> da: 1,0000
27. yeni dnyadaki -> ilk: 1,0000
28. fledii hava -> ile: 1,0000
29. biimindedir bu -> yasaya: 1,0000
30. jones bunun -> nedenini: 1,0000
31. hareket ederler -> scak: 1,0000
32. krlr oysa -> ok: 1,0000
33. gstergesi havann -> snd: 1,0000
34. bilimsel birikim -> olmadan: 1,0000
35. zerine bir -> elektrik: 1,0000
36. yaratc bir -> yntem: 1,0000

37. sylyor termodinamik -> buhar: 1,0000

38. da elektrik -> yk: 1,0000

39. durumdadr oysa -> kalbin: 1,0000

40. tekerlei olan -> bisikletine: 1,0000

41. minnettarm uzun -> yllardan: 1,0000

42. konumaya nerede -> o: 1,0000

43. ilgili baka -> dncelerin: 1,0000

44. koydu cismi -> hzla: 1,0000

45. dnyadaki ilk -> elektrik: 1,0000

46. ile kaplanm -> kayaklarla: 1,0000

47. bir atom -> zincirinin: 1,0000

48. ancak o -> zamanlar: 1,0000

49. gibi bisiklet -> de: 1,0000

50. etkisi olmuyordu -> o: 1,0000

Input:

File Name: BİLİM İŞ BAŞINDA.txt

Total Running Time: 336 millisecond

```
Total Run Time: 336 millisecond
```

Output:

Top 50 items analyzed for 1-gram Analysis:

bir: 541 (0,0346)

ve: 256 (0,0164)

bu: 164 (0,0105)

gregor: 159 (0,0102)

ama: 146 (0,0093)

de: 123 (0,0079)

da: 103 (0,0066)

iin: 98 (0,0063)

gregorun: 95 (0,0061)

gibi: 95 (0,0061)

kzkardei: 93 (0,0059)

daha: 82 (0,0052)

diye: 77 (0,0049)

ne: 75 (0,0048)

babas: 66 (0,0042)

nk: 61 (0,0039)

zerine: 57 (0,0036)

ancak: 56 (0,0036)

sonra: 55 (0,0035)

her: 54 (0,0035)

hi: 53 (0,0034)

pek: 53 (0,0034)

hemen: 53 (0,0034)

o: 51 (0,0033)

kadar: 51 (0,0033)

mdr: 49 (0,0031)

ey: 48 (0,0031)

byle: 46 (0,0029)

btn: 46 (0,0029)

imdi: 45 (0,0029)

yere: 44 (0,0028)

yine: 44 (0,0028)

ok: 43 (0,0028)

annesi: 43 (0,0028)

kzkardeinin: 41 (0,0026)

ki: 40 (0,0026)

biraz: 40 (0,0026)

zerinde: 37 (0,0024)

yle: 36 (0,0023)

en: 35 (0,0022)

bile: 34 (0,0022)

kendini: 34 (0,0022)

anne: 33 (0,0021)

doru: 33 (0,0021)

belki: 33 (0,0021)

onu: 33 (0,0021)

artk: 33 (0,0021)

baka: 31 (0,0020)

babasnn: 31 (0,0020)

kendi: 30 (0,0019)

Top 50 items analyzed for 2-gram Analysis:

1. ksk -> sesle: 1,0000
2. hale -> sokmas: 1,0000
3. zlemle -> aklndan: 1,0000
4. sakn -> koyverme: 1,0000
5. hali -> vard: 1,0000
6. eleen -> gregorla: 1,0000
7. tavandaki -> bu: 1,0000
8. dediin -> uykusunu: 1,0000
9. ykml -> bulunduu: 1,0000
10. silkip -> sarsarak: 1,0000
11. geirmiti -> hi: 1,0000
12. bymt -> hemen: 1,0000
13. kalmadn -> dnmt: 1,0000
14. tiksindii -> yoktu: 1,0000
15. ksz -> hissedecekti: 1,0000
16. srlp -> tuz: 1,0000
17. demeksizin -> doru: 1,0000
18. srlm -> bir: 1,0000
19. tavanda -> daha: 1,0000
20. herkes -> oysa: 1,0000
21. paraland -> bir: 1,0000
22. oynatt -> bu: 1,0000
23. yiyebiliyordu -> genellikle: 1,0000
24. ten -> dilerinin: 1,0000

25. kanayarak -> soluu: 1,0000
26. kesmesine -> karn: 1,0000
27. aznzdan -> anne: 1,0000
28. gideriz -> diye: 1,0000
29. ierde -> asl: 1,0000
30. yanamyor -> robdambr: 1,0000
31. ktn -> anmsad: 1,0000
32. muydun -> annesinin: 1,0000
33. kabartyorlard -> gregor: 1,0000
34. saygya -> davet: 1,0000
35. hakk -> var: 1,0000
36. kurtulmak -> iin: 1,0000
37. blmlerinin -> iinde: 1,0000
38. bykt -> dolaysyla: 1,0000
39. farketmemie -> benziyordu: 1,0000
40. ykld -> yz: 1,0000
41. ykle -> yere: 1,0000
42. girimi -> ancak: 1,0000
43. gereklemeyecek -> yataktan: 1,0000
44. syledi -> nk: 1,0000
45. mdrlne -> bayan: 1,0000
46. istemeyi -> de: 1,0000
47. evrilip -> eve: 1,0000
48. istemeye -> kalkmazd: 1,0000
49. sisin -> grnm: 1,0000
50. aileden -> izin: 1,0000

Top 50 items analyzed for 3-gram Analysis:

1. titreyen omuzlan -> zerinden: 1,0000
2. odasna seirtmi -> annesinin: 1,0000

3. dolduruyor hangi -> yiyecei: 1,0000
4. deitirmeye kendisini -> raz: 1,0000
5. gregor kzkardei -> kilidin: 1,0000
6. yapmadklar bir -> eyi: 1,0000
7. kendisini konservatuara -> yollamay: 1,0000
8. eyleri seip -> seip: 1,0000
9. yekn oluturmutu -> kapnn: 1,0000
10. yaramayacaklarm gibi -> sprgeyle: 1,0000
11. tersine imdiye -> dek: 1,0000
12. aln etkisiyle -> ikide: 1,0000
13. hantal sesi -> iitiliyordu: 1,0000
14. diyerek evresinde -> dolatrd: 1,0000
15. yaatyor ve -> noel: 1,0000
16. kapsnda ayan -> gm: 1,0000
17. durdu sonra -> dnp: 1,0000
18. de payna -> denleri: 1,0000
19. soruturmann ancak -> mdr: 1,0000
20. kestirilemeyen kt -> sonularn: 1,0000
21. yere eilerek -> onun: 1,0000
22. kalmamt artk -> gregor: 1,0000
23. duruma baklrsa -> br: 1,0000
24. dkknnda alp -> kendisine: 1,0000
25. mutfaa geri -> yollanmasnn: 1,0000
26. kapar ve -> holde: 1,0000
27. beraber tastamam -> bir: 1,0000
28. odasyla arasndaki -> uzakln: 1,0000
29. deilse kim -> olduunu: 1,0000
30. gndzn pencereye -> yaklamaktan: 1,0000
31. gregorun arayp -> bulduu: 1,0000
32. neden sonra -> onun: 1,0000
33. denecek biimde -> yapan: 1,0000
34. iyilemi hissediyorum -> nasl: 1,0000
35. resmine iliti -> hemen: 1,0000

36. denedi havada -> debelenen: 1,0000
37. srasnda bazan -> yle: 1,0000
38. sokan kar -> yakasn: 1,0000
39. durumu oktan -> patrona: 1,0000
40. istenmeyen eylerdi -> bunlar: 1,0000
41. yldan beri -> yatp: 1,0000
42. almasna baktn -> kap: 1,0000
43. lacivert bir -> niforma: 1,0000
44. grlt denemezdi -> dmenin: 1,0000
45. gelen babasyd -> ne: 1,0000
46. rasgele yenen -> berbat: 1,0000
47. nlenemeyen sla -> benzer: 1,0000
48. hep akamlan -> uruyor: 1,0000
49. hole gelebilir -> ve: 1,0000
50. iliti gz -> grd: 1,0000

Input:

File Name: BOZKIRDA.txt

Total Running Time: 341 millisecond

```
Total Run Time: 341 millisecond
```

Output:

Top 50 items analyzed for 1-gram Analysis:

bir: 623 (0,0360)
diye: 208 (0,0120)
ne: 200 (0,0116)
de: 157 (0,0091)
bu: 152 (0,0088)
yakov: 145 (0,0084)
dedi: 137 (0,0079)
da: 129 (0,0075)
vasili: 126 (0,0073)

malva: 121 (0,0070)

ve: 120 (0,0069)

gibi: 96 (0,0055)

ben: 94 (0,0054)

sonra: 90 (0,0052)

sen: 72 (0,0042)

o: 71 (0,0041)

daha: 70 (0,0040)

seryojka: 70 (0,0040)

iin: 69 (0,0040)

ki: 64 (0,0037)

mi: 59 (0,0034)

kadar: 57 (0,0033)

ey: 57 (0,0033)

ok: 57 (0,0033)

fakat: 55 (0,0032)

m: 54 (0,0031)

ama: 54 (0,0031)

bana: 52 (0,0030)

var: 51 (0,0029)

ona: 50 (0,0029)

kadn: 49 (0,0028)

sesle: 48 (0,0028)

sordu: 48 (0,0028)

her: 47 (0,0027)

onun: 46 (0,0027)

yle: 46 (0,0027)

zaman: 45 (0,0026)

ha: 43 (0,0025)

yok: 41 (0,0024)

onu: 40 (0,0023)

bak: 39 (0,0023)

imdi: 38 (0,0022)

sana: 38 (0,0022)

ya: 37 (0,0021)

seni: 37 (0,0021)

doru: 35 (0,0020)

ar: 35 (0,0020)

bunu: 35 (0,0020)

hi: 34 (0,0020)

zerinde: 33 (0,0019)

Top 50 items analyzed for 2-gram Analysis:

1. nianlmdr -> bay: 1,0000
2. srme -> zamannn: 1,0000
3. yinelemesi -> zek: 1,0000
4. hale -> gelen: 1,0000
5. zlemle -> syledii: 1,0000
6. sarktm -> slak: 1,0000
7. dediin -> ey: 1,0000
8. fingirdemek -> iin: 1,0000
9. kst -> sakince: 1,0000
10. isterdim -> dorusu: 1,0000
11. oynatm -> malva: 1,0000
12. zamankinden -> daha: 1,0000
13. yapsn -> szerken: 1,0000
14. yakmak -> iin: 1,0000
15. kta -> oturmu: 1,0000
16. gzelliinden -> eser: 1,0000
17. cierlerim -> taze: 1,0000
18. attrmaya -> balamt: 1,0000
19. kte -> oturdu: 1,0000
20. balksn -> ne: 1,0000
21. davranlacan -> bilirim: 1,0000
22. yklr -> gider: 1,0000

23. ter -> iinde: 1,0000
24. muydun -> seryojka: 1,0000
25. ktr -> gne: 1,0000
26. hakl -> buluyordu: 1,0000
27. kurtulmak -> isteyen: 1,0000
28. yklm -> rengi: 1,0000
29. evremize -> yyordu: 1,0000
30. grmeye -> balamt: 1,0000
31. vermeye -> de: 1,0000
32. kaslmt -> bu: 1,0000
33. kua -> koltuk: 1,0000
34. gllerin -> gszler: 1,0000
35. haberin -> yoktur: 1,0000
36. cevapларn -> yazmaya: 1,0000
37. byklar -> arasndan: 1,0000
38. vururdu -> sonra: 1,0000
39. kul -> kle: 1,0000
40. skyor -> srt: 1,0000
41. efkati -> daha: 1,0000
42. ucundan -> tamam: 1,0000
43. roln -> ballandra: 1,0000
44. olmaktan -> tr: 1,0000
45. zavalls -> ne: 1,0000
46. avucumdaki -> son: 1,0000
47. glyorsun -> niye: 1,0000
48. svyor -> bir: 1,0000
49. sama -> ey: 1,0000
50. moskova -> niversitesinin: 1,0000

Top 50 items analyzed for 3-gram Analysis:

1. malva anszn -> glmeye: 1,0000
2. o tandk -> siyah: 1,0000

3. olurum zellikle -> de: 1,0000

4. siyah finosunu -> grdn: 1,0000

5. ktn malva -> da: 1,0000

6. ya bir -> diyeceim: 1,0000

7. okunuyordu ceketinin -> n: 1,0000

8. kime olacak -> bolese: 1,0000

9. o namussuzda -> ya: 1,0000

10. dnd ar -> ar: 1,0000

11. vedalat ky -> boyunca: 1,0000

12. balklar bordadan -> trmanarak: 1,0000

13. ha diye -> hrldad: 1,0000

14. ilgi gren -> bu: 1,0000

15. vaadini bir -> kez: 1,0000

16. vard iinde -> ki: 1,0000

17. der gibi -> byle: 1,0000

18. deil ah -> sensin: 1,0000

19. misin hayr -> yleyse: 1,0000

20. parasdr buralardaysa -> fingirdemek: 1,0000

21. brakalm bunlar -> bu: 1,0000

22. bakanlnca yaynlanmam -> ancak: 1,0000

23. kk sandal -> yan: 1,0000

24. sevgilim meryem -> anamz: 1,0000

25. geebilirsin fakat -> imdi: 1,0000

26. kenetleyerek uzanm -> gkyzn: 1,0000

27. dilinde daha -> dorusu: 1,0000

28. sonra bitkin -> bir: 1,0000

29. anszn telalanarak -> barakaya: 1,0000

30. karlk vermek -> istedii: 1,0000

31. iitildi znk -> diye: 1,0000

32. yldzlar titrek -> titrek: 1,0000

33. mi yok -> aslanm: 1,0000

34. eh anan -> unutma: 1,0000

35. karanlkta gri -> bir: 1,0000

36. tanrm tanrm -> yaral: 1,0000
37. duyuyordu onun -> karsnda: 1,0000
38. kadnn gnln -> yapmak: 1,0000
39. grntler belirdi -> renci: 1,0000
40. mutlaka ekmei -> de: 1,0000
41. dalgann zerinde -> hzla: 1,0000
42. beklemeden birbiri -> arkasna: 1,0000
43. buradasn ay -> geti: 1,0000
44. ne bulursanz -> aln: 1,0000
45. zene bezene -> kaleme: 1,0000
46. denizden delikanl -> deniz: 1,0000
47. birdenbire denizden -> kp: 1,0000
48. dik olarak -> dtkleri: 1,0000
49. parltlardan aka -> okunuyordu: 1,0000
50. bouk ve -> ksa: 1,0000

Input:

File Name: grimms-fairy-tales_P1.txt

Total Running Time: 264 millisecond

```
Total Run Time: 264 millisecond
```

Output:

Top 50 items analyzed for 1-gram Analysis:

the: 747 (0,0726)
and: 546 (0,0531)
to: 271 (0,0264)
a: 244 (0,0237)
he: 233 (0,0227)
his: 142 (0,0138)
in: 141 (0,0137)
was: 139 (0,0135)
of: 127 (0,0124)

it: 124 (0,0121)

you: 114 (0,0111)

that: 109 (0,0106)

i: 100 (0,0097)

said: 95 (0,0092)

as: 93 (0,0090)

but: 93 (0,0090)

him: 90 (0,0088)

they: 88 (0,0086)

so: 86 (0,0084)

had: 85 (0,0083)

for: 83 (0,0081)

with: 80 (0,0078)

on: 78 (0,0076)

all: 73 (0,0071)

at: 71 (0,0069)

her: 69 (0,0067)

not: 63 (0,0061)

then: 58 (0,0056)

be: 56 (0,0054)

will: 56 (0,0054)

came: 53 (0,0052)

she: 51 (0,0050)

my: 51 (0,0050)

went: 46 (0,0045)

when: 46 (0,0045)

into: 44 (0,0043)

have: 42 (0,0041)

out: 42 (0,0041)

good: 39 (0,0038)

is: 39 (0,0038)

what: 38 (0,0037)

very: 38 (0,0037)

one: 38 (0,0037)

up: 38 (0,0037)

this: 38 (0,0037)

by: 37 (0,0036)

old: 37 (0,0036)

there: 36 (0,0035)

away: 35 (0,0034)

could: 35 (0,0034)

Top 50 items analyzed for 2-gram Analysis:

1. half -> of: 1,0000
2. staying -> here: 1,0000
3. hall -> now: 1,0000
4. pretend -> to: 1,0000
5. music -> the: 1,0000
6. saves -> shoeleather: 1,0000
7. understanding -> this: 1,0000
8. because -> i: 1,0000
9. coals -> and: 1,0000
10. stabbed -> him: 1,0000
11. ashamed -> of: 1,0000
12. thus -> everything: 1,0000
13. chimneys -> could: 1,0000
14. leapt -> down: 1,0000
15. impetuous -> disposition: 1,0000
16. compassionate -> heart: 1,0000
17. willow -> spray: 1,0000
18. courtyard -> and: 1,0000
19. stop -> and: 1,0000
20. squared -> his: 1,0000
21. espying -> the: 1,0000
22. beware -> of: 1,0000

23. weeks -> old: 1,0000
24. pockethandkerchief -> put: 1,0000
25. least -> they: 1,0000
26. youre -> right: 1,0000
27. gold -> and: 1,0000
28. hungry -> too: 1,0000
29. cudgel -> laid: 1,0000
30. quiet -> they: 1,0000
31. stump -> of: 1,0000
32. staircase -> ending: 1,0000
33. combed -> his: 1,0000
34. secretly -> to: 1,0000
35. toes -> cracked: 1,0000
36. breathed -> her: 1,0000
37. weeping -> then: 1,0000
38. prettier -> than: 1,0000
39. mumbled -> something: 1,0000
40. destroyed -> all: 1,0000
41. calling -> to: 1,0000
42. spin -> but: 1,0000
43. robber -> ran: 1,0000
44. thank -> me: 1,0000
45. roaring -> out: 1,0000
46. itbut -> where: 1,0000
47. thingyou -> will: 1,0000
48. mewed -> and: 1,0000
49. dragging -> one: 1,0000
50. whats -> the: 1,0000

Top 50 items analyzed for 3-gram Analysis:

1. man with -> a: 1,0000
2. bath and -> gave: 1,0000

3. should like -> to: 1,0000
4. only want -> a: 1,0000
5. the day -> long: 1,0000
6. singer and -> may: 1,0000
7. her likewise -> if: 1,0000
8. quarters that -> they: 1,0000
9. without asking -> her: 1,0000
10. wind took -> himself: 1,0000
11. of making -> an: 1,0000
12. time forward -> sultan: 1,0000
13. as she -> had: 1,0000
14. sitting in -> the: 1,0000
15. bird in -> this: 1,0000
16. bird is -> i: 1,0000
17. door sprang -> open: 1,0000
18. had come -> and: 1,0000
19. again when -> any: 1,0000
20. years hans -> in: 1,0000
21. met them -> it: 1,0000
22. silver helped -> hans: 1,0000
23. brook as -> he: 1,0000
24. walls the -> spit: 1,0000
25. thieves dont -> care: 1,0000
26. that there -> he: 1,0000
27. away went -> the: 1,0000
28. golden one -> that: 1,0000
29. sleeping briar -> rose: 1,0000
30. put her -> into: 1,0000
31. so loud -> that: 1,0000
32. riding is -> no: 1,0000
33. it sinking -> in: 1,0000
34. boy the -> box: 1,0000
35. fox but -> he: 1,0000

36. queen very -> much: 1,0000
37. goosegrease for -> six: 1,0000
38. village and -> employed: 1,0000
39. bean thanked -> him: 1,0000
40. still holding -> up: 1,0000
41. still more -> and: 1,0000
42. from whence -> do: 1,0000
43. only get -> in: 1,0000
44. of pure -> gold: 1,0000
45. for wear -> i: 1,0000
46. soon grew -> round: 1,0000
47. the wheelbarrow -> drove: 1,0000
48. should come -> to: 1,0000
49. king could -> not: 1,0000
50. him we -> will: 1,0000